

Subject Description Form

Subject Code	EE549
Subject Title	Modern Sensor Technologies
Credit Value	3
Level	5
Pre-requisite/ Co-requisite/ Exclusion	Undergraduate-level circuit and electromagnetic theory
Objectives	1. To acquire the fundamentals of advanced sensor technologies. 2. To make the students to understand the structures and working principles of resistive, capacitive, piezoelectric, acoustic, electric and magnetic sensors. 3. To enable the students to understand and design thermal and mechanical sensors, optical sensors, optical fiber sensors and micro-electromechanical system (MEMS) sensor technologies. 4. To know the applications of sensors in various industrial applications.
Intended Learning Outcomes	Upon completion of the subject, students will be able to: a. Acquire the operation principles and recent developments of sensors and transducer technologies, including thermal and mechanical sensors, electric and magnetic sensors, optical sensors as well as MEMS sensors technologies. b. Understand the structures and working principles of thermal sensors, mechanical sensors, acoustic sensors, electric and magnetic sensors for practical applications. c. Select the most appropriate optoelectronic components and optical fiber devices to design optical sensors and optical fiber sensor systems. d. Comprehend the structures and multidisciplinary working principles of MEMS-technology and sensor networks. e. Have hands-on experience in the assembling and testing of electric/optical sensors or MEMS sensors.
Subject Synopsis/ Indicative Syllabus	1. Introduction to sensor fundamentals. Definition of sensors; sensor and information; physical quantities; relation between quantities; sensor classification; uncertainty aspects. 2. Thermal, mechanical and acoustic sensors. Thermoresistive and thermoelectric sensors; construction, general properties and applications of force sensors, accelerometer, pressure sensors, velocity sensors, and inertial sensors; microphones, ultrasonic sensors and their applications. 3. Electric and magnetic sensors. Magnetic induction, permeability and magnetostriction; electric and magnetic field sensor; Hall effect and magnetometers; voltage and current sensors and applications. 4. Optical sensors and optical fiber sensors. Classification of optical sensors; quantum-based optical sensors; photoelectric sensors; charge-coupled device (CCD) sensors; optical fiber interferometer sensors; optical fiber grating sensors and applications. 5. MEMS and optical MEMS sensors. Production of MEMS; MEMS-based pressure sensors, accelerometers, hot-wire anemometry and gyroscopes; optical MEMS sensors. 6. Applications of sensor technologies. Sensor technologies for power systems; sensor technologies for wind turbines; sensor technologies in smart railway

	network and systems. Laboratory Experiments: Testing and calibration of force sensors and on-board MEMS accelerometers.						
Teaching/Learning Methodology	Lectures, quizzes, tests, laboratory experiments, mini projects, and examination.						
	Teaching/Learning Methodology	Outcomes					
		a	b	c	d	e	
	Lectures	√	√	√	√		
	Tutorials	√	√	√	√		
	Experiments/Mini-project	√		√		√	
Assessment Methods in Alignment with Intended Learning Outcomes	Specific assessment methods/tasks	% weighting	Intended subject learning outcomes to be assessed				
			a	b	c	d	e
	1.Tests/Quizzes	18%	√	√	√	√	
	2. Assignments	6%	√	√	√	√	
	3. Lab and mini project	16%	√		√		√
	4. Examination	60%	√	√	√	√	
	Total	100%					
	This subject introduces the structures, working principles and applications of electrical/optical sensor technologies. Tests/assignments/examination will be used to assess the outcomes about the structures and operation principles and applications of various electrical/magnetic/optical sensors. Experiments/mini project will be used to assess the hands-on experience in electrical/optical sensors and MEMS devices.						
	Student Study Effort Expected	Class contact:					
		▪ Lectures/Tutorials/Laboratory demo				39 Hrs.	
Other student study effort:							
▪ Mini-project and report				20 Hrs.			
▪ Self-study and assignments				46 Hrs.			
Total student study effort				105 Hrs.			
Reading List and References	1. Sensors for Mechatronics, 2 nd edition, Paul P. L Regtien, Edwin Dertien, Elsevier, 2018. 2. Sensors, actuators, and their interfaces: a multidisciplinary introduction, Nathan Ida, SciTech Publishing, 2014. 3. Handbook of Modern Sensors: Physics, Designs, and Applications, Jacob Fraden, Springer International Publishing AG, 2015. 4. Sensors handbook, 2 nd edition, Sabrie Soloman, McGraw-Hill, 2010.						